

INTRODUCTION

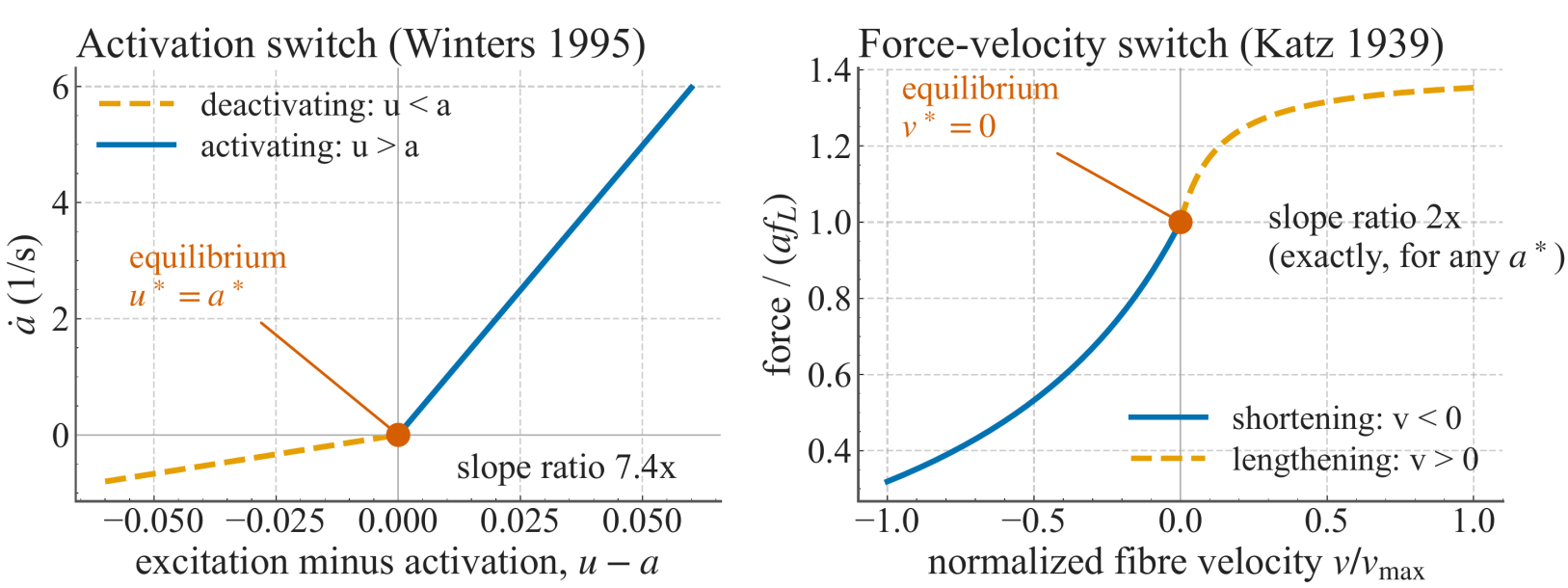
One equilibrium. Four directional linear models.

Muscle-driven robots, prostheses and exosuits often use controllers designed around an equilibrium linearization. This assumes that a single local model describes small perturbations.

The Hill-type formulation studied here switches twice: activation dynamics changes when excitation crosses activation, and the force–velocity relation changes between shortening and lengthening.

At equilibrium, excitation equals activation and velocity is zero. Both switching surfaces therefore pass through the operating point.

Question: how much does the choice of branch affect prediction and closed-loop control?



Activation slopes differ by 7.45× at the reference equilibrium; velocity slopes differ by 2×.

METHODS

A simulated elbow against gravity

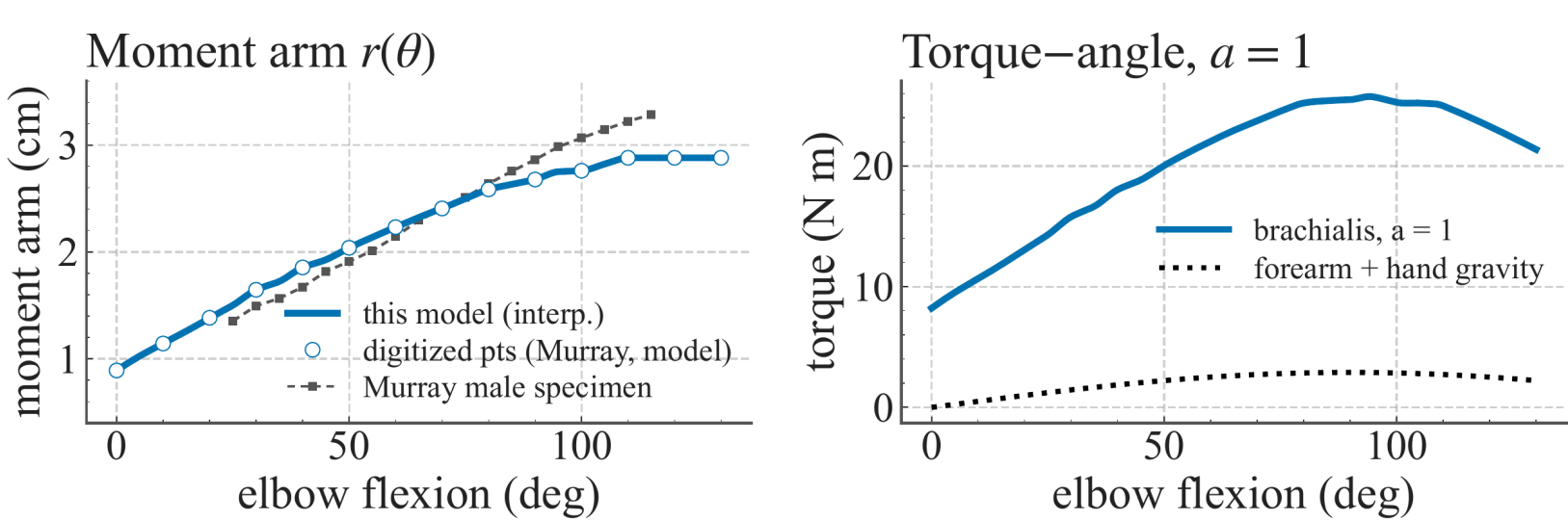
A forearm–hand pendulum is driven by a rigid-tendon Hill-type brachialis model. Muscle parameters follow Holzbaur; activation dynamics follows Thelen; moment arms are digitized from Murray et al.

Reference posture: 60°
Equilibrium activation: 0.113

Prediction test: apply a small +0.01 excitation step and compare all four local models with the nonlinear trajectory.

Feedback test: linear MPC with a 10 ms sample time and 20-step horizon. Keep controller weights fixed; vary only the branch and linearization point.

Extensions: held-load equilibria, a brachialis–triceps antagonist pair, noise, plant mismatch and delay.



Model checks: moment arm and torque–angle relationship. Brachialis peaks at 25.8 N m near 94°.

RESULTS

14–75% prediction error from a wrong branch

At the reference posture, matched-branch error stays near 1% of the true angular swing. Mismatched models diverge within 50–100 ms.

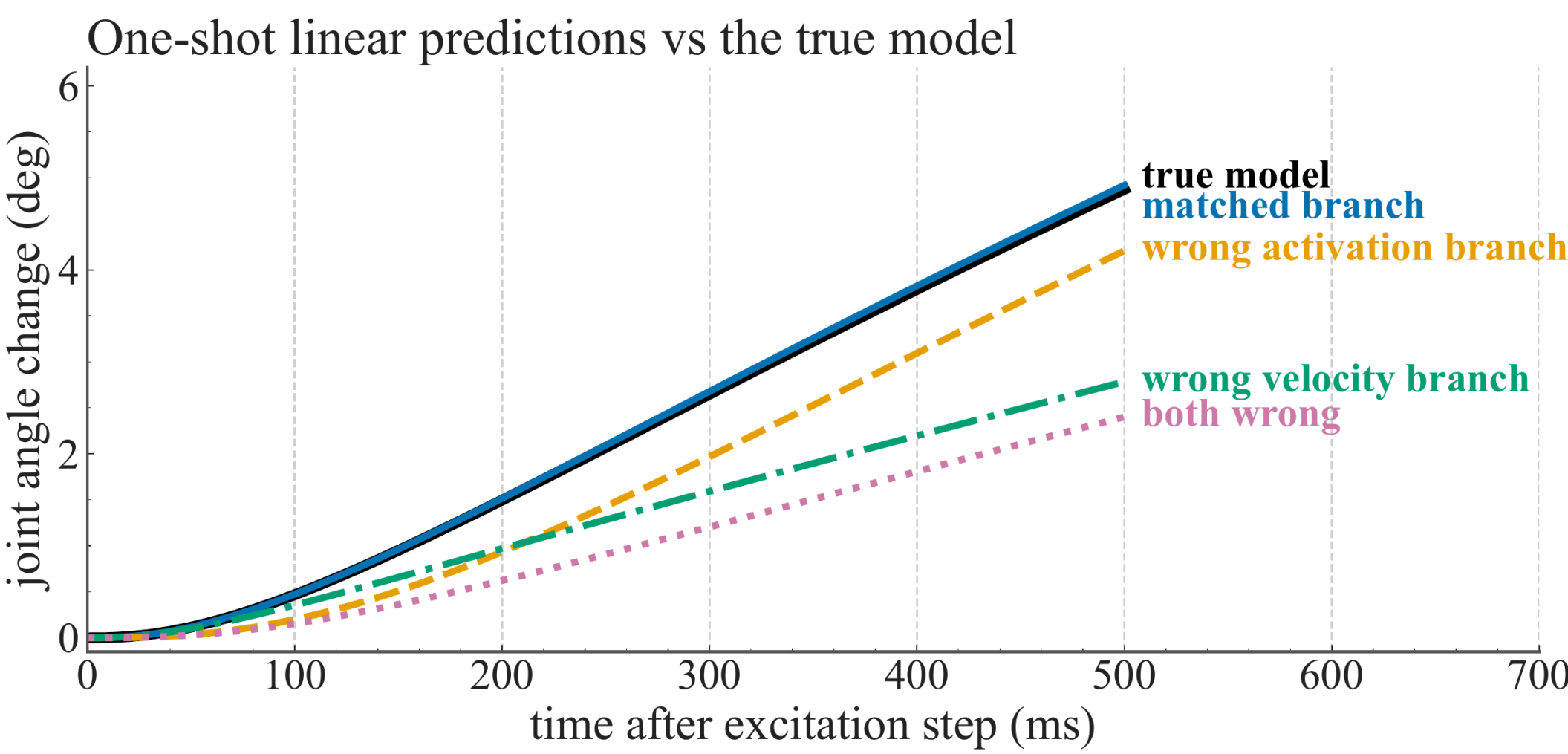
At 50 ms: wrong activation branch, 71%; wrong velocity branch, 14%; both wrong, 75%. These are errors relative to the true swing, not absolute angles.

The effect persists across operating points

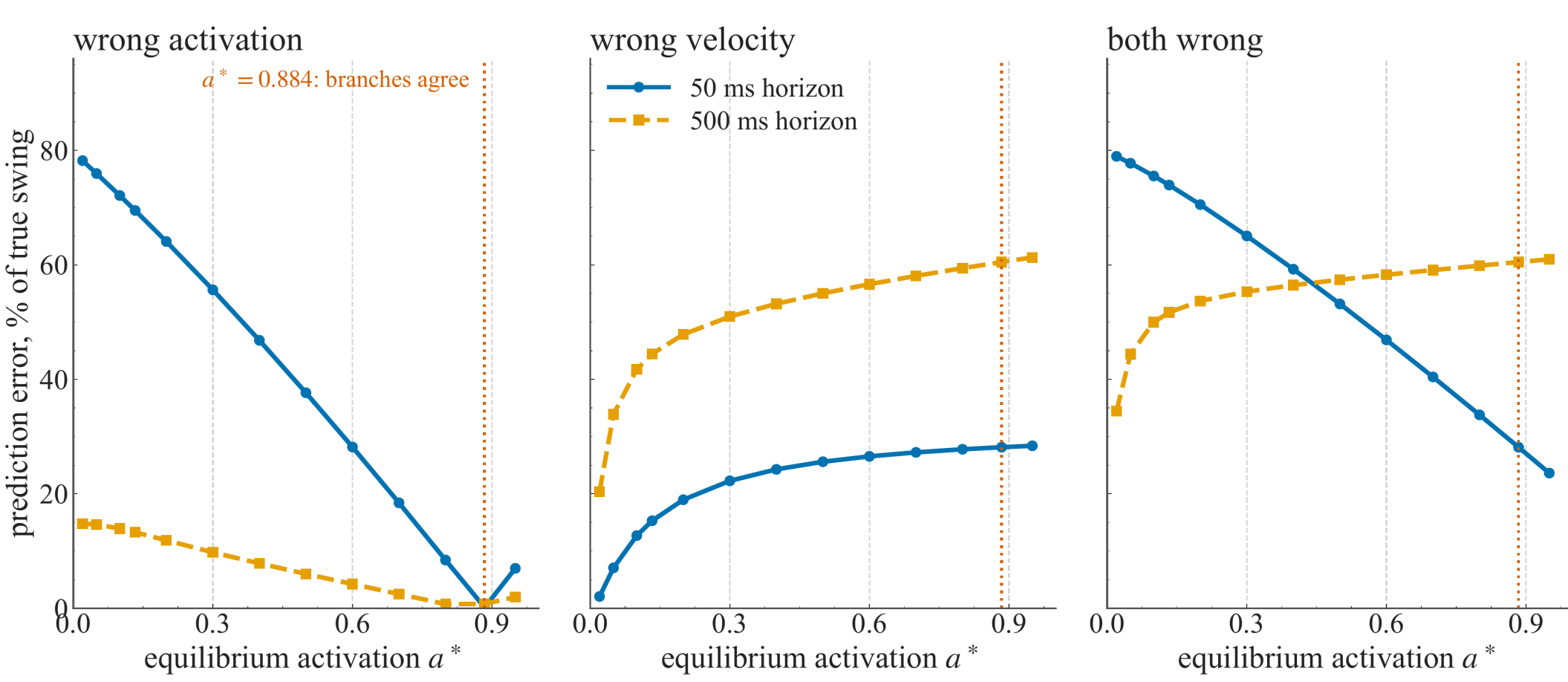
With both branches wrong, error is at least 23% of the activating-step signal at every single-muscle equilibrium tested.

Feedback changes the consequence

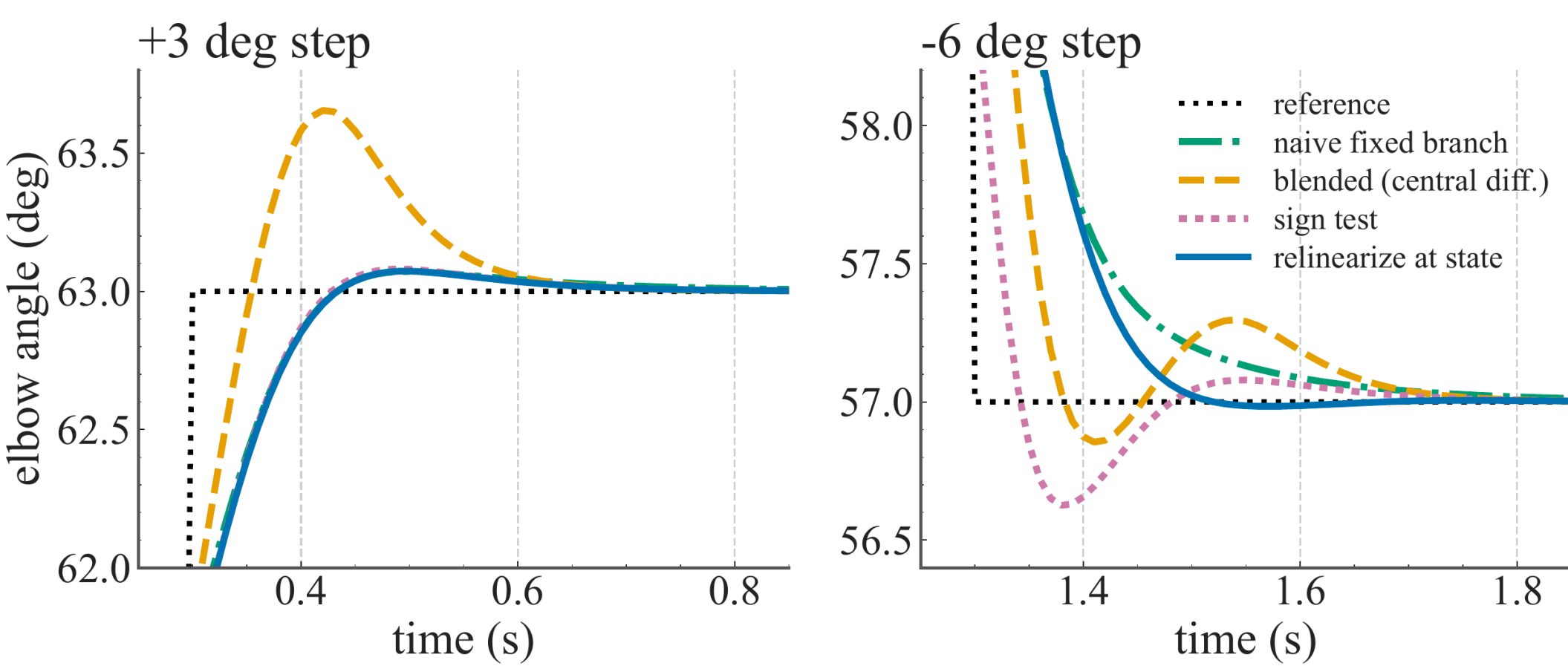
In the tested delay-free cases, feedback preserves stability and steady-state tracking, but differences in transient quality remain. Blending the branches gives 8–9× more raising overshoot than the fixed, sign-test and relinearized designs.



A small excitation step: the matched branch tracks the nonlinear model; wrong branches diverge.



Held-load sweep: activation-branch error falls as activation rises; velocity-branch error grows.



Closed-loop steps: 60° → 63° → 57°. Relinearization reduces lowering-step overshoot.

Read together: branch choice affects prediction across operating points and remains visible in closed-loop transients.

DISCUSSION

Relinearize at the measured state

During lowering, braking re-activates the muscle. A branch chosen only from the initial sign misses this mid-plan change.

Lowering overshoot falls from 0.37° with the sign test to 0.02° with relinearization.

Smoothing activation retains much of the transient cost. Tracking rankings depend on weights and noise.

A predictor restored stability under the tested 20 and 50 ms delays.

Baseline MPC: overshoot and tracking error (degrees).

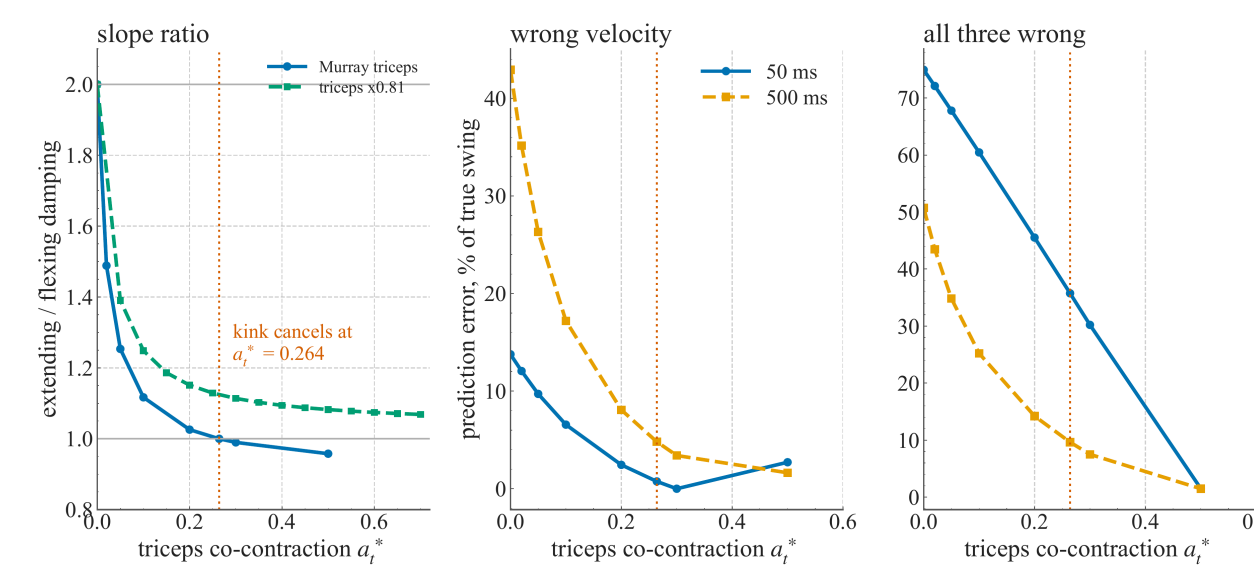
Model	Up	Down	Sine RMS
Blended	0.65	0.15	0.196
Smooth act.	0.42	0.48	0.169
Sign test	0.08	0.37	0.153
Relinearize	0.07	0.02	0.149

CONCLUSIONS

Choose the branch as well as the operating point.

Use directional local models and relinearize as the state changes. Co-contraction can reduce the velocity kink, but adds an activation switch and effort.

Scope: one simulated joint. Hardware validation, reflexes and history-dependent muscle effects remain open.



Antagonist co-contraction reduces the velocity kink toward cancellation.

REFERENCES

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CONTACT & MATERIALS

Alan Royce Gabriel Samuel

bs22b001@smail.iitm.ac.in

Poster source and figures:
github.com/alanroyce2010/hill_IROS

Simulation results from the existing project draft.